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Cognitive Psychology

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I, Rhea Madhogarhia, pledge that the work on this examination is completely my own. I have not received assistance from anyone or from ChatGPT/GPT4/other LLMs to complete this examination at any time. I also pledge that I will not provide assistance to any other students, at any time, before, during or after this examination. I realize I can use my textbook, material from the canvas site, and my notes. I realize that if I am found to break this pledge, I will receive a 0 on this final.

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Part I.

Note: My citations are listed in the answer option descriptions rather than in the problem statements.

Question 1 (Weeks 1-3)

Cognitivism sparked a paradigmatic revolution, the cognitive revolution, in psychology. What fundamental principle in pre-cognitivism psychology, did the cognitive revolution *primarily* challenge?

a. Cognitivism aimed to reject the reliance on introspective methods of understanding consciousness and cognitive processes.
b. Cognitivism disputed the idea that observable behavior should be included in our study of psychological processes and, instead, sought to focus only on the significance of mental processes and the study of the mind.
c. Cognitivism aimed to challenge the dominance of behaviorism in psychological study because of how behaviorisms severely limited the study of cognition through the exclusion of the mind in psychological inquiry.
d. Cognitivism took issue with Tolman's findings that behavior is strengthened through positive reinforcement in his experiments on rats and posited that operant conditioning could not be explained through behavioral observation.

Explanation:

This question hopes that the student recognizes that the cognitive revolution aimed to reintroduce the study of the mind into psychology after Watson's founding of behaviorism rejected that consciousness was a viable area of study in psychology (pg. 10, textbook). The key word in the question is "primarily" and hints that some answers may be partially correct but not the best answer to the question.

Option A is misleading because it correctly questions the reliability of introspective methods - a problem with structuralism/introspectionism, the approach to studying the mind that came before behaviorism (pg 8, textbook). While cognitivism acknowledges that introspection cannot be the only source of information/evidence for studying the mind, the shift to cognitivism was not sparked by the issues with structuralism. Option B is closer to the reason for the shift because it mentions behaviorism's focus on observable behavior dictating psychological research, but cognitivism did not promote the exclusion of behavioral observation in psychological study (pg. 10, textbook). This leads us to the correct answer, option C. Option C is correct because cognitivism disputed the *dominance* of behaviorism (pg. 13, textbook). In other words, cognitivism aimed to bring the study of the mind, peering into the "black box", back into psychological conversations, complementing behavioral observations (Slides 59, 78, W1L1). Option D correctly identifies Tolman's findings as important in the development of cognitivism but misrepresents the findings of the experiment and confuses Tolman's work with Skinner's (pg. 11-12, textbook). Skinner's experiments on operant conditioning were important in the study of behaviorism and a cognitivist would argue that operant conditioning does not fully substantiate the rats' behavior, but a cognitivist would also not discount the same behavioral observations Skinner made as useful information in the study of the rats' minds.

Question 2 (Weeks 4-6)

In our textbook's chapter on attention we discussed inattentional deafness and the cocktail party effect. What is the main distinction between the cocktail party effect and inattentional deafness?

e. The cocktail party effect demonstrates our ability to attend to one stimulus while ignoring others - selective attention-, but inattentional deafness refers to how we can choose to drown out auditory stimuli when specifically focused on visual tasks.
f. The cocktail party effect deals with the unconscious processing of relevant auditory information whereas inattentional deafness refers to our ability to tune out irrelevant auditory stimuli.
g. The cocktail party effect explains our ability to shift our attention between multiple auditory stimuli, but inattentional deafness highlights how we fail to recognize auditory stimuli while focused on a visual task.
h. The cocktail party effect refers to our ability to process auditory stimuli amidst other competing auditory stimuli whereas inattentional deafness involves a failure to notice auditory stimuli due to high load visual tasks.

Explanation:

This question aims to have the student recognize the difference between inattention and selective attention. Also, Lavie's load theory of attention is implicated in both of these phenomena: for the cocktail party, a person's ability to selectively attend will ultimately be affected by the person's cognitive load, and, for inattentional deafness, load theory helps explain how we can be actively attending and unintentionally lack the perception of auditory stimuli because our attentional load is already greatly occupied (pg. 99-101, textbook; Slide 3, 26 W5L2; Slide 49,50, W5L1).

Option E correctly identifies selective attention as the driver behind the cocktail party effect but incorrectly defines inattentional deafness as something we can choose to do (pg. 95, textbook). We do not choose to miss or drown out auditory stimuli while focusing on a visual task - we miss perceiving auditory stimuli because of our filled attentional load (Slide 26, W5L2). Option F touches on a caveat of the cocktail party effect instead of focusing on its actual definition: unconscious processing of relevant information could be hearing your name at a party while engrossed in a conversation you are having with your friend (Slide 9, W6L1). Once again, this option mischaracterizes inattentional deafness as something we can choose to do. Option G talks about shifting attention rather than selectively attending with regard to the cocktail party effect but correctly characterizes inattentional deafness. Option H is the correct answer because it provides correct definitions of both the cocktail party effect and inattentional deafness and correctly identifies that they both tell us something about attention's role in processing auditory stimuli (pg. 96, 116-117, textbook).

Question 3 (Weeks 7-9)

Which of these neurological observations are associated with insight problem solving as opposed to analytical problem solving?

i. Left hemispheric activation both temporally and spatially shown in EEG and fMRI scans during tasks that require semantic integration.
j. Occipital alpha band EEG activity spikes right before insights make their way into consciousness
k. A sudden and heightened activation of the anterior cingulate before arriving at a solution born out of insight
l. Enhanced right hemispheric and diffuse visual cortex activity during resting state

Explanation:

This question revolves around week 9's core reading, "The Cognitive Neuroscience of Insight", by Kounios and Beeman. To answer this question, students must think about the different neurological events that occur before and during insight problem solving. Many of the

answers are misleading because they confuse either spatial or temporal information from Kounios and Beeman's findings.

Option I is a trap because it points out the hemispheric asymmetry in activation in insight problem solving and also recognizes the dual EEG and fMRI approach described in the paper (pg. 79, Kounios & Beeman). The primary flag in this option to signal that it is incorrect should be "left". It is right, not left, hemispheric activation that is shown during semantic integration and processing, a processing and manipulation vital to insight (pg. 82, Kounios & Beeman). Option J is correct and touches on the sudden appearance of insight into consciousness by using the word "spike" rather than something like "gradually rise". This finding is referenced by the "Attending In, Out, and Around" section of Kounios and Beeman's paper (p. 83). While the occipital alpha activation is sudden, the anterior cingulate's activation is not fully tied to a solution to a problem arriving into consciousness. Option K is therefore incorrect because the greater activation of the anterior cingulate is more involved with exploring nondominant solution possibilities (pg. 86, Kounios & Beeman). Option L is incorrect because enhanced right hemispheric and diffuse activity during resting state in the visual cortex is associated more with a person's susceptibility for creative and insightful problem solving than it is actually having an insight. These findings point towards insightful thinkers having a hemispheric bias but that isn't to say that a person with this enhanced resting state activation may not also solve a problem analytically (81-82, Kounios & Beeman).

*next page has part 2 essay

Part II

"Funes the Memorious", a fictitious short story by Jorge Luis Borges, tells the story of a man unable to categorize things in the world around him: if light shines on a blade of grass causing it to shine a new color, Funes is unable to recognize and categorize that newly colored blade of grass as the same blade he just laid eyes on moments earlier. In lecture, we discussed how, in this story, Funes was hypothetically experiencing the world directly, rather than through a mental representation or a world informed by concepts. Because of this, Funes is unable to make inferences about the world around him - his ability to retrieve old memories, form hypotheses, generalize or summarize events or situations, or even recognize people in a new pair of glasses is impaired. In remembering absolutely everything as a new concept or object, Funes' life arguably becomes much more difficult and complex to navigate (Slide 56, W8L1). Our everyday lives are shaped by our ability to call on prior knowledge, and create new conceptual knowledge in ways that are manageable for our minds - rationing cognitive resources, being wary of cognitive load. Our expectations, derived from prior knowledge and experience, coupled with our ability to categorize information and direct attention, serve as pivotal mechanisms that inform our behavior and streamline our navigation of the world. These cognitive processes intricately shape how we interpret, respond to, and interact with our environment, allowing a more fulfilling engagement with the ever changing and complex world around us.

We cannot possibly interact with every stimuli presented in our environment. Helping us avoid cognitive and sensory overload when interacting with others and our environment, our ability to attend selectively to pertinent information (deemed pertinent by expectations and prior

knowledge), allows us to manage our cognitive load and navigate our environments effectively. Selective attention refers to the process in which we focus on specific features in our environment while filtering out distractions or irrelevant information. What we deem as irrelevant or relevant is influenced by our expectations and knowledge: a study was conducted to see if radiologists would notice that photos of gorillas were superimposed onto brain scans (Slide 27, W5L1). The radiologists were not expecting to find tumors in the region that the gorilla was placed and therefore selectively attended to areas of the scan that they expected a tumor to be found in. The results of this experiment found that many radiologists did not notice the gorillas on the 3D lung scans - because they were selectively attending to another part of the scan, they succumbed to inattentional blindness (Slide 21, W5L1). In lecture we discussed that this did not make the radiologists bad doctors. If anything, they were so attentive to their job, finding tumors, that any irrelevant information did not distract them. This ability to filter out task-irrelevant stimuli is seen to be a product of the load theory of attention proposed by Lavie (pg. 99, textbook). The radiologists were performing an arguably high load task, searching for tumors in a brain scan, and therefore there was little attentional capacity left to notice things like a gorilla in a place that was already left out in the radiologists' selective attendance.

Attention is also instrumental in order to "experience a coherent world", as mentioned by our textbook (pg. 119). Attention is an important factor in perception and can help us bind the features of an object to create our perception of it. We can look at the effect this has on our ability to recognize objects by discussing Treisman's feature integration theory (FIT) and an individual suffering from Balint's syndrome (pg. 121 textbook). Briefly, FIT says that objects are broken into parts and those parts are categorized into features of the original object; then, the features are combined through focused attention and this informs our perception of an object.

Patient R.M. had parietal lobe damage that took away his ability to focus attention on individual objects - Balint's syndrome. As a result, patient R.M was unable to correctly combine and later ascribe an object's features to the object - a red T and a blue O became a red O or a blue T. Treisman's experiments on her own theory also showed that prior knowledge was important for categorizing and binding features of an object during the focused attention stage: in an illusory conjunction experiment, similar to the one performed on patient R.MTreisman told the participants what they were being shown. If she said the word "carrot", the participants would perceive the upside down orange triangle to be a carrot. Fitting the shapes' features to a familiar category allowed them to improve on the task and correctly identify the objects when prompted (pg. 121, textbook)

Revisiting Jorge Luis Borges' character, Funes, it's interesting to think about how his inability to use prior knowledge or make assumptions about objects cause Funes to have a nearly perfect memory, a byproduct of his condition that Professor Heald mentioned in class. Are imperfect or incomplete memories a good thing? One way to think about this "incomplete" memory is to think about how some actors learn their lines for a play. Personally, I have a friend who learns the first letter of every word in their script. Once they've memorized that, they write it down and begin to remember what each letter stood for in a sentence - similar to a very long acronym. This demonstrates chunking - a way of increasing the amount of information we are able to store in our limited capacity memory (pg. 140, textbook; Slide 13, W6L2). Once the long acronym was memorized, the retrieval of this acronym then brought the memory into working memory where my friend could manipulate the information stored in the acronym to represent the words in the script he was trying to memorize. This memory was incomplete in the sense that he didn't memorize every word of the script, but it did allow him to remember his lines faster

and, keeping in mind that memory has a limited capacity, more efficiently. While this story was anecdotal and doesn't exactly demonstrate turning small bits of information into more meaningful units, it showed that we don't need a perfect memory to remember the things we want to remember. It would simply take a lot of work and rehearsal to just memorize a bunch of words or digits on a page, but we have processes by which this rehearsal and encoding can become easier - prior knowledge and expertise make storing lots of information in memory a little bit easier. Ericsson conducted a digit memory study in which a patient, SF, was asked to use chunking to recode digits into chunks that would make it easier for him to store in his memory. SF's chunking was improved by using his expertise as a runner to reprogram arbitrary digits he was asked to memorize into running times. He retrieved prior knowledge from his long term memory to increase the capacity of his short term memory system to hold up to 79 digits at a time which he could then retrieve and decode in working and short term memory (pg. 141, textbook). We also see expertise awarding memory advantages to experts in their fields; chess masters are significantly better at memorizing playable non random chess boards because they are able to chunk based on prior knowledge of the game and configurations of pieces (Slide 17, W6L2). Chess masters are therefore able to solve problems in a game much more quickly and efficiently than novices are because of their ability to retrieve information from long term stores and use that information to chunk information effectively into their short term memory (pg. 375, textbook). The core reading, "The Mind and Brain of Short-Term Memory", also touches on how chunked information can be maintained in short term memory through the focus of attention by retrieval from LTM or perceptual encoding (pg. 212). Attention and prior knowledge constantly intertwine to direct our perception, memory and other cognitive processes.

Revisiting our discussion of chess masters, the chess masters performed just as poorly as the novices when board configurations were randomized. They weren't able to call on prior knowledge anymore which made producing a solution to their task just as difficult to find as for anyone else. When the correct knowledge cannot be applied, solutions become harder to find. The idea that prior knowledge can help direct and guide one's attention towards the right place, like in the gorilla brain scan experiment, and that solutions are found more easily when the correct knowledge is applied, helps us understand how insights in problem solving can be found (Slide 11, W9L2). In lecture, insight was described as the process of activating the appropriate knowledge for a certain problem. For example, in Weisberg and Alba's nine dot experiment, it is not until participants are told they can draw lines outside of the box that the task of connecting the dots is no longer confined to the shape the dots made (snapping out of functional fixedness). Only with the right knowledge were a substantial, though not majority, amount of people able to solve the problem (Slide 20, W9L2). SImilarly, in the out of focus pictures task, the pictures look just like noise if you are given no context. As soon as someone is told that there is an elephant somewhere in the image, the viewer begins to reframe their thinking and allow their mind to entertain new possibilities for the image. While some may look for a tusk, big ears, and four feet, others may suddenly see the whole elephant, all at once, after altering their expectation of what the image contains (Slide 18, W9L2). While expectation can be a good thing for problem solving, some consequences of knowledge and expectation, mental set and functional fixedness, can actually make problem solving harder (pg. 360, textbook). In these cases, attentional release can spark creativity and divergent thinking. This can be done through a myriad of methods - taking a walk, meditating, showering, sleeping, and mind wandering (pg. 386-389, textbook). While mind wandering can seem counterproductive to problem solving there is a neurological

evidence that there is a connection between mind wandering and creativity facilitated by the default mode network (pg. 384-385, textbook). It's hard to forget what you know, because sometimes it's all you can hang on to, but that release of attentional control can let your mind reframe its view and go where you didn't know you needed it!